ARCHIVAL TECHNICAL MONOGRAPH // SERIES MM-2026-P

ATELIER // PRESENCE

A Low-Latency, Privacy-Preserving Multimodal Telemetry and Pedagogical Equilibrium Architecture for Physical and Hybrid Learning Spaces

SYSTEM ARCHITECTURE SPECIFICATION

Core Daemon	FastAPI Async Engine (Python 3.11 Runtime)
Computer Vision	OpenCV Haar Frontal + Euclidean Centroid Bipartite Matcher
Acoustic DSP	WebAudio 16kHz PCM + Librosa Centroid (300-3400Hz)
Persistence Engine	SQLite 3.4x Write-Ahead Logging (WAL Mode, Sync=Normal)
Pedagogy Layer	Hybrid Groq Llama 3.1 8B + Gemini 3.5 Flash + 6-State Offline

COMPUTATIONAL PRIVACY GUARANTEES

Biometric Storage	**Zero Face Persistence** · Decoded RAM frames purged per tick
Tracking Vector	Dimensionless Centroids (cX, cY) in memory only
Verification Gate	10.0s Integral Presence Verification · 3.0s Occlusion Grace
Throttling Shield	10.0s Calibrated Telemetry Cache · ≤ 120 tokens JSON
Data Export	RFC 4180 Streaming CSV via Pandas DataFrame Engine

ZERO BIOMETRIC RETENTION · IN-MEMORY CENTROID LOOP (<14MS) · SQLITE WAL CONCURRENCY · 16-BAND FFT EQUALIZER · RATE-PROTECTED PEDAGOGICAL AI

ARCHIVAL MONOGRAPH ABSTRACT // PHILOSOPHICAL FOUNDATIONS

REF: SPEC-2026-PRESENCE-V4

Institutional monitoring systems have historically approached classroom environments with surveillance paradigms that undermine student psychological safety, introduce brittle cloud dependencies, and accumulate sensitive biometric liability. *Atelier // Presence* treats presence telemetry not as surveillance, but as a tactile, ambient pedagogical mirror rooted in computational sovereignty.

By decoupling presence verification from facial identification, the architecture relies on mathematical Euclidean centroid tracking, temporal integral presence state machines, and real-time speech formant filtering (300Hz-3400Hz). Coupled with an air-gapped SQLite WAL database and a rate-protected pedagogical advisor, the system sustains a continuous cognitive equilibrium—fostering deep focus, human rhythm, and pedagogical clarity.

PARADIGM & SECURITY FOUNDATIONS

01 / System Paradigm & Security Architecture

Computational sovereignty through non-identifiable sensory abstraction, zero biometric storage, and SQLite WAL persistence.

1. ZERO BIOMETRIC PERSISTENCE

Incoming base64 camera frames are decoded directly into ephemeral NumPy RAM buffers. No facial embeddings, deep templates, or raw pixel tensors are ever committed to disk or transmitted to third-party endpoints.

2. LOCAL CENTROID TRACKING

Detected faces are abstracted immediately into 2D spatial centroids (cX, cY). Gaze vectors and velocity dynamics reside exclusively in RAM within the `TrackedParticipant` lifecycle and evaporate upon session termination.

3. AIR-GAPPED SQLITE WAL

Telemetry sessions, verified attendance entries, and pedagogical records persist into a local SQLite database utilizing Write-Ahead Logging (`WAL`), achieving concurrent, lock-free streaming writes and instant CSV exports.

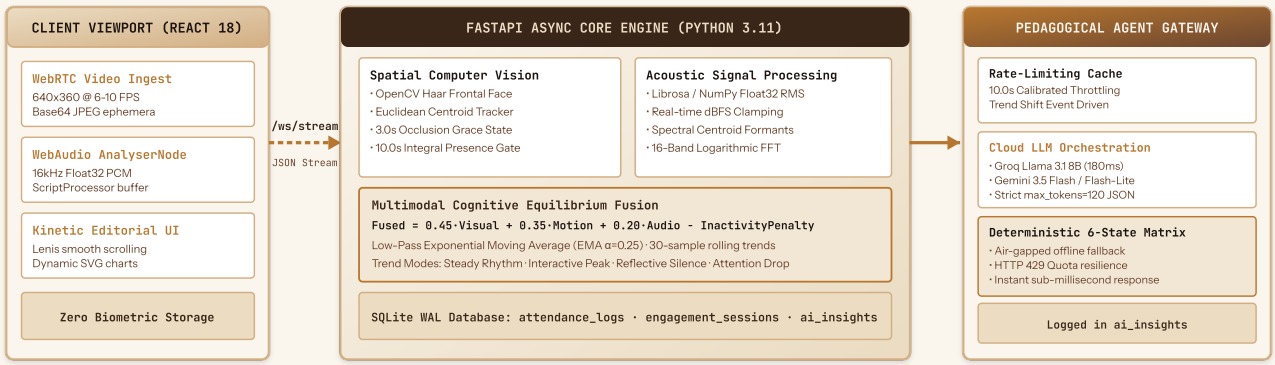


FIGURE 1.1: SYSTEM TOPOLOGY, IN-MEMORY PROCESSING BOUNDARIES & AIR-GAPPED SQLITE PERSISTENCE

§ 1.1 In-Memory Ephemeral Gating

The architecture strictly enforces an **ephemeral memory lifecycle**. High-resolution WebRTC frames are decoded in Python memory via OpenCV's C-bindings. As soon as spatial coordinates (x, y, w, h) are mapped to dimensionless centroids (cX, cY), the pixel buffer is garbage-collected.

Participants are designated with session-scoped pseudonymized identifiers (e.g. `Participant_01`). If a participant leaves the sensory aperture for more than 3.0 seconds, the corresponding track instance is purged, guaranteeing zero retrospective biometric re-identification.

§ 1.2 SQLite WAL Mode Concurrency

High-frequency telemetry logging creates significant I/O pressure in traditional file locks. *Atelier // Presence* configures SQLite with **Write-Ahead Logging**:

> **`PRAGMA journal_mode=WAL; PRAGMA synchronous=NORMAL;`**
>
> Decouples concurrent reader queries from the append-only write loop.

A dedicated shared-memory index (`.db-shm`) coordinates reader snapshots against the WAL buffer (`.db-wal`). Administrative users can stream real-time CSV analytics without blocking the 10 FPS telemetry daemon.

SPATIAL TELEMETRY & ATTENDANCE VERIFICATION

02 / Computer Vision & Spatial Centroid Pipeline

Multi-scale Haar feature detection, Euclidean distance matrix bipartite gating, and the 10.0s sustained presence state machine.

CENTROID SPATIAL PROJECTION

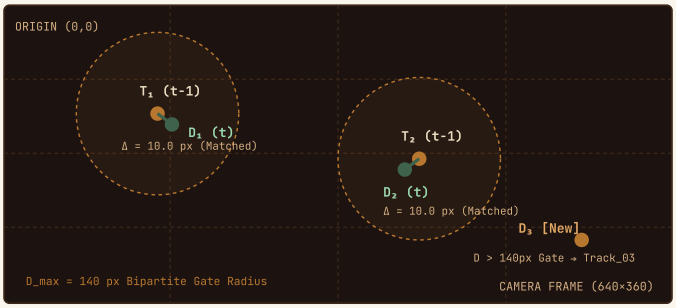
For detected bounding boxes $B_k = (x_k, y_k, w_k, h_k)$ from the Haar Cascade classifier, geometric center mass points (cX_k, cY_k) are calculated:

$$cX_k = x_k + \frac{w_k}{2}, \quad cY_k = y_k + \frac{h_k}{2}$$

PAIRWISE EUCLIDEAN DISTANCE MATRIX

Between active tracks $i \in \{1 \dots M\}$ and incoming detections $j \in \{1 \dots N\}$, the bipartite distance matrix is evaluated:

$$D_{i,j} = \sqrt{(cX_{\text{track}, i} - cX_{\text{det}, j})^2 + (cY_{\text{track}, i} - cY_{\text{det}, j})^2}$$



BIPARTITE EUCLIDEAN ASSIGNMENT MATRIX

	D₁ (Det 1)	D₂ (Det 2)	D₃ (Det 3)	Action
T₁ (Trk 1	10.0 px*	147.1 px	246.8 px	Pair T₁→D₁
T₂ (Trk 2)	138.4 px	10.0 px*	102.9 px	Pair T₂→D₂

• Greedy Argmin Pairing Strategy:
Sorted row-minimum distances are assigned subject to gating limit D ≤ 140px.
• Detection D₃ remains unassigned to existing tracks: instantiated as new Track_03.
• Deterministic bipartite matching executes in O(MN) time under 0.8ms.

FIGURE 2.1: CENTROID MAPPING IN CAMERA COORDINATES & DISTANCE MATRIX BIPARTITE GATING

§ 2.1 3.0s Occlusion Grace State Machine

Studio environments naturally involve student head turns, notebook sketching, and temporary book occlusions. Punitive trackers lose tracking state immediately when a face is occluded for a single frame, resetting participant tenure.

Atelier // Presence tracks consecutive missing duration:

$$\Delta t_{\text{disappeared}} = t_{\text{curr}} - t_{\text{last_seen}}$$

If $\Delta t_{\text{disappeared}} \le 3.0\text{ s}$, the track remains active in memory. Deregistration occurs strictly when $\Delta t > 3.0\text{ s}$.

Additionally, brisk hallway walk-bys through open studio doors are filtered via low-pass horizontal velocity gating: $|v_x| = |(cX_t - cX_{t-1}) / \Delta t| > 65\text{ px/s}$.

§ 2.2 10.0s Sustained Presence Integral

To eliminate transient false positives from passing visitors, physical attendance follows a strict temporal integration lifecycle:

$$\text{Duration}(T_i) = t_{\text{curr}} - t_{\text{first_seen}}$$

DURATION WINDOW	VERIFICATION STATE	LEDGER COMMIT
$0.0\text{ s} \le t < 3.0\text{ s}$	TRANSIENT (<3S)	Ephemeral RAM only
$3.0\text{ s} \le t < 10.0\text{ s}$	VERIFYING...	Telemetry stream only
$t \ge 10.0\text{ s}$	VERIFIED PRESENT	**Committed to SQLite**

Upon crossing the 10.0-second threshold, `_auto_commit_verified_attendance()` executes an atomic SQLite transaction, logging the participant entry.

HEAD POSE ORIENTATION & SOMATIC DYNAMICS

03 / Head Pose & Optical Motion Inactivity Engine

Dual-engine solvePnP/geometric fallback, angular cosine attention decay, and normalized 64×64 ROI frame differencing.

DUAL-ENGINE HEAD POSE ARCHITECTURE

1. MediaPipe FaceMesh (Primary): Extracts 468 landmarks. Six canonical 3D model coordinates (Nose tip, Chin, Left/Right Eye corners, Left/Right Mouth corners) are mapped via `cv2.solvePnP` with iterative Levenberg-Marquardt optimization. Rodrigues conversion yields Euler angles $(\theta_y, \theta_p, \theta_r)$.

2. Mathematical Geometric Fallback (Air-Gap): When MediaPipe is uncompiled, estimates orientation from optical axis offsets and bilateral cheek luminance asymmetry:

$$H_{\text{asym}} = \frac{\mu_{\text{left}} - \mu_{\text{right}}}{\mu_{\text{left}} + \mu_{\text{right}}}, \quad \text{Yaw} = 55.0 \cdot H_{\text{asym}} + 12.0 \cdot x_{\text{norm}}$$

ANGULAR COSINE ATTENTION DECAY

Natural human attention does not collapse precipitously at small head rotations. *Atelier // Presence* models gaze orientation through a continuous cosine decay:

$$\text{Score}(\theta) = \begin{cases} 100.0 & \theta \le 15^\circ \\ 25.0 + 37.5 \left(1 + \cos\left(\pi \frac{\theta - 15}{30}\right)\right) & 15^\circ < \theta \le 45^\circ \\ 25.0 & \theta > 45^\circ \end{cases}$$

Where $\theta = \max(|\theta_y|, |\theta_p|)$. Protected note-taking floor: $\ge 70.0\%$.

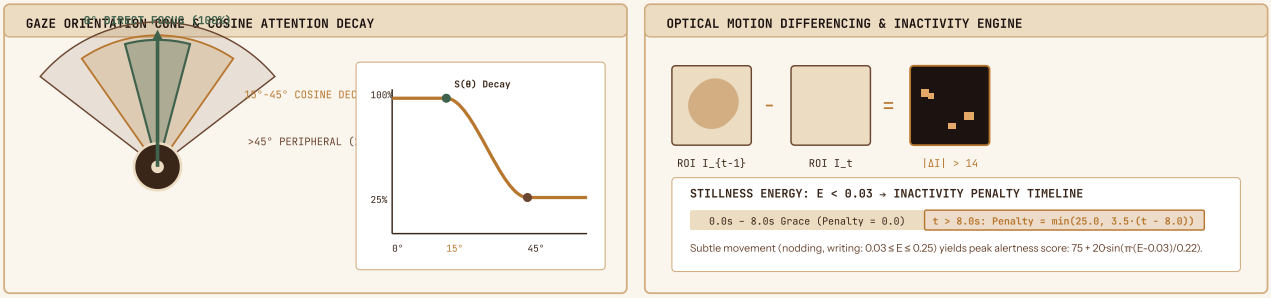


FIGURE 3.1: 3D HEAD POSE ORIENTATION CONES, COSINE ATTENTION DECAY, AND MOTION ENERGY PENALTIES

§ 3.1 Note-Taking & Downward Gaze Protection

Standard computer vision systems routinely penalize participants who glance down at their physical notebooks, misclassifying scholarly drafting as inattention. *Atelier // Presence* detects downwards tilt by computing the facial vertical luminance gradient:

$$R_{TB} = \frac{\mu_{\text{forehead}}}{\mu_{\text{chin}}}, \quad \text{is_note_taking} = (dy > 4.5\text{ px}) \lor (R_{TB} < 0.77)$$

When active, the participant is classified as `Note-Taking / Downward Focus`, and the attention score is clamped to a healthy pedagogical baseline ($\ge 70\%$).

§ 3.2 ROI Optical Flow Differencing

Each tracked face ROI is scaled to a standard 64×64 grid and filtered with a Gaussian kernel ($\sigma=1.0$). Absolute differencing isolates microscopic motor gestures:

$$E_{\text{ROI}} = \frac{1}{64 \times 64} \sum_{x,y} \mathbb{I}(|I_t(x,y) - I_{t-1}(x,y)| > 14)$$

Continuous postural stagnation for over 8.0 seconds accumulates an inactivity penalty, gently lowering overall equilibrium and alerting the pedagogical agent to recommend a posture break.

ACOUSTIC SPECTROMETRY & DSP PIPELINE

04 / Acoustic Intelligence & WebAudio DSP Pipeline

WebAudio AnalyserNode PCM streaming, Librosa RMS/dBFS energy, and the 16-band logarithmic spectral equalizer.

ROOT-MEAN-SQUARE (RMS)

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^N x_i^2}$$

Real-time energy over $N=512$ float32 PCM samples in $[-1.0, 1.0]$.

DECIBELS FULL SCALE (DBFS)

$$\text{dBFS} = 20\log_{10}(\max(\text{RMS}, 10^{-6}))$$

Calibrated silence threshold set at -42.0 dBFS floor.

SPECTRAL CENTROID (F_C)

$$f_c = \frac{\sum_k f_k |X(k)|}{\sum_k |X(k)|}$$

Center of mass discriminating voice from ambient HVAC noise.

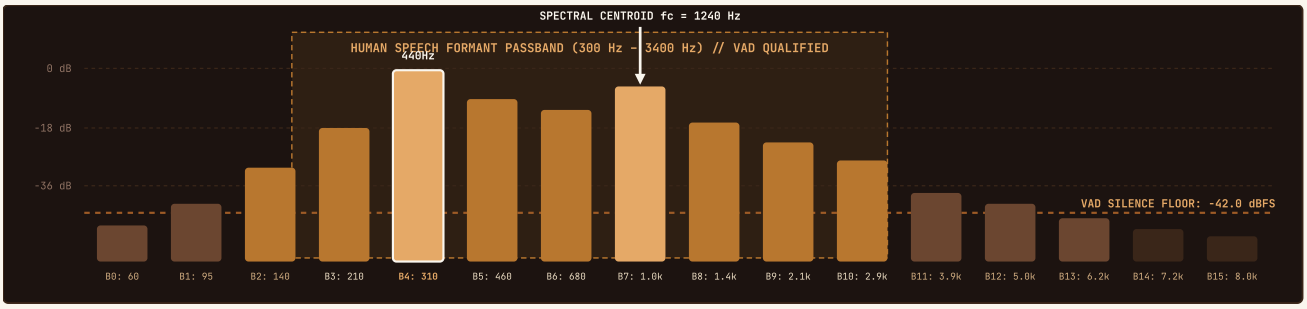


FIGURE 4.1: 16-BAND LOGARITHMIC SPECTRAL EQUALIZER & SPEECH FORMANT PASSBAND (CONCERT A PEAK AT BIN 4)

§ 4.1 Frequency Band Specifications

BAND	FREQUENCY	ACOUSTIC ROLE
Bin 0–1	60 Hz – 140 Hz	HVAC ambient rumble & low sub-bass floor
Bin 2–3	140 Hz – 310 Hz	Male & female fundamental vocal pitch (F_0)
Bin 4	310 Hz – 460 Hz	**Vowel body (440 Hz Concert A resonance)**
Bin 5–6	460 Hz – 1000 Hz	Formant F_1 low/high vowels ('o', 'u', 'a')
Bin 7–8	1000 Hz – 2100 Hz	Vocal clarity & Formant F_2 ('e', 'i')
Bin 9–10	2100 Hz – 3900 Hz	Projection, Formant F_3, speech intelligibility
Bin 11–12	3900 Hz – 6200 Hz	Consonant attack & high sibilance ('s', 'sh')
Bin 13–15	6200 Hz – 9500 Hz	Studio air shimmer & room reflections

§ 4.2 Speech Formant Discrimination

Unlike naive volume meters that spike when a door slams or paper rustles, the *AudioAnalyzer* evaluates acoustic qualification across two orthogonal axes:

$$\text{is_speech} = (\text{dBFS} > -42.0\text{ dBFS}) \land (300\text{ Hz} \le f_c \le 3400\text{ Hz})$$

True speech requires both sufficient decibel energy and spectral centroid alignment.

When active speech is detected, the acoustic score feeds into the Multimodal Equilibrium Formula at 20% weight, signaling participatory studio dialogue.

FUSION MATHEMATICS & PEDAGOGICAL DECISION SUPPORT

05 / Multimodal Cognitive Equilibrium & Advisory AI

Fused Equilibrium Formula, low-pass Exponential Moving Average ($\alpha=0.25$), and rate-protected LLM pacing.

FUSED COGNITIVE EQUILIBRIUM FORMULATION

EXPONENTIAL SMOOTHING A = 0.25

$$\text{Fused Score}_{\text{raw}} = 0.45 \cdot \text{Visual Attention} + 0.35 \cdot \text{Movement Score} + 0.20 \cdot \text{Acoustic Score} - \text{Inactivity Penalty}$$

$$\text{EMA}_t = 0.25 \cdot \text{Fused Score}_{\text{raw}, t} + 0.75 \cdot \text{EMA}_{t-1}$$

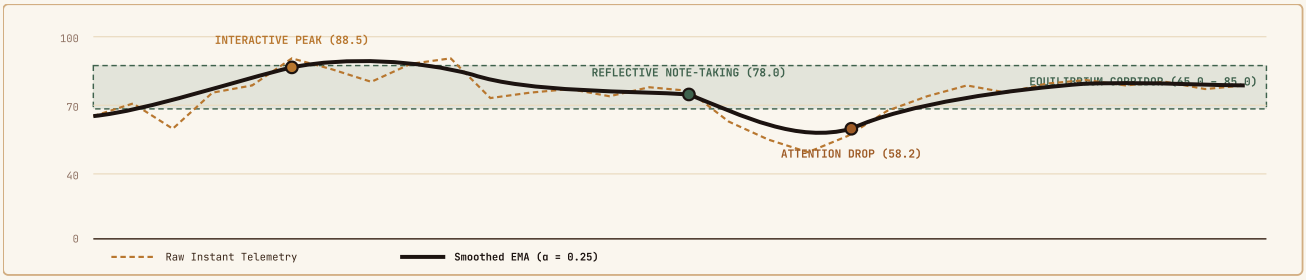


FIGURE 5.1: 30-SAMPLE MULTIMODAL TELEMETRY FUSION & EXPONENTIAL MOVING AVERAGE (EMA) CURVE

§ 5.1 Rate-Protected Agentic AI Gateway

Pedagogical recommendations are synthesized via cloud LLMs without exposing telemetry streams to rate exhaustion:

- **Groq Llama 3.1 8B Instant:** Ultra-low latency (~180ms), JSON output schema.
- **Gemini 3.5 Flash / Flash-Lite:** Deep multimodal reasoning with thinking budget capped.
- **10-Second Calibrated Throttling Cache:** External LLM calls fire *only* if the engagement trend shifts or after 10.0s elapsed cooldown.
- **Output Cap:** Strictly enforced at `max_tokens = 120`.

§ 5.2 Deterministic 6-State Offline Matrix

If network connectivity fails or API rate quotas are exceeded (HTTP 429), the engine defaults instantly to the deterministic matrix:

TRIGGER CONDITION	PEDAGOGICAL ACTION
Active Faces = 0	Awaiting Cohort Arrival
Visual Attention < 50%	Address Gaze Drift & Fatigue
Inactivity Penalty ≥ 5.0	Prompt Posture Shift / Break
Speech Active & Audio ≥ 70%	Facilitate Peer Dialogue
Fused Score ≥ 75%	Deepen Theoretical Complexity
Baseline Steady State	Sustain Current Narrative Rhythm

DATA SCHEMAS & PROTOCOL SPECIFICATIONS

06 / Data Engineering, Persistence & API Reference

SQLite relational schema, Write-Ahead Logging concurrency, WebSocket streaming contract, and RFC 4180 CSV export.

SQLITE AIR-GAPPED RELATIONAL SCHEMA (WAL CONCURRENT)

`PRAGMA JOURNAL_MODE=WAL; PRAGMA SYNCHRONOUS=NORMAL;`

```
-- 1. Verified & In-Progress Participant Presence Ledger
CREATE TABLE attendance_logs (
    id INTEGER PRIMARY KEY AUTOINCREMENT, session_id VARCHAR(64) INDEX, participant_id VARCHAR(64) INDEX,
    first_seen DATETIME, last_seen DATETIME, timestamp DATETIME INDEX, sustained_seconds FLOAT DEFAULT 0.0,
    status VARCHAR(32) DEFAULT 'Tracking...' -- 'Verified Present', 'Verifying...', 'Transient'
);

-- 2. Granular Time-Series Telemetry & Sensory Feature Records
CREATE TABLE engagement_sessions (
    id INTEGER PRIMARY KEY AUTOINCREMENT, session_id VARCHAR(64) INDEX, timestamp DATETIME INDEX,
    active_faces INTEGER, head_pose_score FLOAT, movement_score FLOAT, inactivity_penalty FLOAT,
    audio_rms_score FLOAT, fused_score FLOAT, trend VARCHAR(64)
);

-- 3. Pedagogical Pacing & Decision Insights Journal
CREATE TABLE ai_insights (
    id INTEGER PRIMARY KEY AUTOINCREMENT, session_id VARCHAR(64) INDEX, timestamp DATETIME INDEX,
    engagement_context VARCHAR(255), suggestion_text VARCHAR(1024), pacing_action VARCHAR(128), provider VARCHAR(64)
);
```

RESTFUL TELEMETRY INTERFACE & PANDAS CSV STREAM

ENDPOINT	METHOD	FUNCTION & PROTOCOL ROLE	RESPONSE TYPE
/api/health	GET	Sensory daemon health check, database status & WAL state	application/json
/api/attendance	GET	Chronological ledger of verified attendees sorted by timestamp	application/json (list)
/api/analytics/trends	GET	Aggregated session metrics (mean fused score, peak active faces)	application/json
/api/calibration	GET/POST	Dynamic recalibration of sustained threshold (s), VAD floor (dB)	application/json
/api/export/csv	GET	RFC 4180 streaming CSV download via Pandas DataFrame engine	text/csv; charset=utf-8
/ws/stream	WS	Full-duplex real-time camera & audio telemetry stream (6–10 FPS)	Bidirectional JSON

CLIENT WS PAYLOAD FORMAT

```
{
  "frame": "...",
  "audio_samples": [-0.012, 0.045, 0.082],
  "mic_rms": 0.045,
  "simulate": false
}
```

SERVER WS RESPONSE FORMAT

```
{
  "annotated_frame": "data:image/jpeg;base64...",
  "participants": [{ "id": "Participant_01", "status": "Verified Present", "att
  "metrics": { "fused_score": 82.5, "trend": "Steady Rhythm", "is_speech": true
  "ai_suggestion": { "suggestion": "Deepen complexity...", "provider": "Gemini
}
```

ATELIER // PRESENCE · ARCHITECTURAL TREATISE MONOGRAPH

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